

E-Commerce Database Description

This document describes the database design of an e-commerce system. The database is structured to support user management, product catalog, order processing, payment handling, and additional features such as reviews and notifications.

1. USERS

Stores user information including authentication and profile data.

- 1 Register and login
- 2 Manage profile
- 3 Link to orders and carts

2. ADDRESSES

Stores multiple shipping addresses for each user.

- 1 Used in order checkout
- 2 Supports default address selection

3. CATEGORIES

Organizes products into hierarchical categories.

- 1 Supports filtering and navigation
- 2 Parent-child relationship

4. PRODUCTS

Main product information stored here.

- 1 Linked to category and brand
- 2 Used for display and purchase

5. PRODUCT_VARIANTS

Stores variations like size, color, version.

- 1 Stock management
- 2 Different pricing per variant

6. PRODUCT_IMAGES

Stores images related to products.

- 1 Primary and secondary images
- 2 Ordered display

7. CARTS

Represents a user's shopping cart.

- 1 Each user has one cart
- 2 Temporary storage before checkout

8. CART_ITEMS

Products inside the cart.

- 1 Stores quantity and variant
- 2 Linked to products

9. ORDERS

Stores order details.

- 1 Contains total price and status
- 2 Linked to user and address

10. ORDER_ITEMS

Detailed products inside an order.

- 1 Snapshot of product at purchase time
- 2 Prevents data inconsistency

11. PAYMENTS

Stores payment transaction details.

- 1 Tracks payment status
- 2 Linked to orders

12. PAYMENT_METHODS

Available payment methods.

- 1 COD, Banking, E-wallet

13. VOUCHERS

Discount codes applied to orders.

- 1 Percentage or fixed discount
- 2 Validity control

14. SHIPMENTS

Tracks delivery process.

- 1 Tracking code
- 2 Delivery status

15. PRODUCT_REVIEWS

User reviews for products.

- 1 Rating and comments
- 2 Verified purchase

16. WISHLISTS

User's saved products.

- 1 Bookmark products
- 2 User preference tracking

17. NOTIFICATIONS

System notifications.

- 1 Order updates
- 2 Promotions

18. SEARCH_HISTORIES

Stores search activity.

1 Improves recommendation system